

Harrison Grant

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Skills

- **Fabrication:** 3D printing (FDM, SLA, SLS), 2D cutting (water jet, laser cutter), woodworking, metal machining (manual and CNC), sheet metal construction, additive and subtractive DFM, soldering
- **Software:** Solidworks, OnShape, Fusion 360, Java, C, MATLAB, Cura, PreForm, Linux, TrueNAS

Education

Boston University College of Engineering

Bachelor of Science in Mechanical Engineering

Boston, MA

Expected May 2026

- 3.8 GPA, Dean's List
- Coursework: CAD, Thermodynamics, Differential Equations, Product Design, Manufacturing, Dynamics

Professional Experience

JM Electrical - Experts in commercial electrical and automation systems

Project Management Intern

Lynnfield, MA

Summer 2025

- Collaborated with project managers to Assemble estimates and project quotes by analyzing scope documents and technical drawings
- Compared equipment submittals and client specifications to ensure alignment with project requirements

RASTIC - Boston University's premier robotics lab

Research Assistant and Lab Manager

Boston, MA

Spring 2025

- Built out lab infrastructure, designed tools including self contained pneumatics box for actuating soft robotics
- Provided guidance on CAD, mechanical design, and custom electronics to students supporting their projects

Formlabs Inc - Innovators in additive manufacturing

Research and Development Intern

Somerville, MA

Summer 2024

- Performed design for manufacturing (DFM) optimizations to make complicated geometry manufacturable with in house tools, saving time and cost versus outsourcing
- Designed and ordered sheet metal parts to assemble bespoke filter box integrating with air circulation system
- Ideated, built, and implemented testing rig to validate functionality of mission-critical components

Inventia Life Science - Series B-funded bioprinting startup

Engineering Intern

Sydney, NSW

Spring 2024

- Designed and executed series of experiments to evaluate multiple bio-printer accuracy metrics simultaneously
- Improved image-processing workflow by creating spreadsheet to aggregate data en masse and expose trends
- Presented to senior managers, engineering team about hardware fixes to address axis-level imprecision

Pursuit Aerospace - Manufacturer of complex aircraft components

Manufacturing Intern

Malden, MA

Summer 2023

- Designed, and produced custom protective cover for engine intake to protect parts during manufacturing
- Identified systematic issues causing major defects in important product line

Collaborative Projects

Boston University Gliding and Soaring Club - Founding member and executive board

- Trained in wing removal of Stemme S6 motor-glider including control surface safety and preflight checks
- Designed multi-part center console plate to prevent wire fatigue and extend operational length of components
- Drew template to water jet cut carbon fiber sheet and created system for interchangeable 3D printed inserts

Chessinator - Chess Playing Robot

- Repurposed 3D printer driver board, stepper motors, and aluminum extrusion, making robot capable of moving chess pieces anywhere around a 12" acrylic board
- Integrated servo and electromagnet with 3D printer control board to pick and place pieces

Coin Collecting Robot and Hackathon Victor

- Modeled 3D environment to approximate actual competition conditions, enabling more precise CAD
- Built robot capable of picking, storing, and placing coins in small containers, earning 2nd place at competition

Laser Communication System

- Improvised binary encoding scheme and wrote code to efficiently encode and send five digit messages
- Created complete CAD model of a fully 3D printable turntable powered by continuous servo motor

BU Terrier Motorsport - Electric Racing Club (Society of Automotive Engineers)

- Took initiative to learn lathe and manual mill operation in order to fabricate parts
- Collaborated during technical discussions to offer unseen solutions to roadblocks reached by team members